

Internal innovation disclosure – not a filed patent application  
Generated entirely from real, already-computed pipeline outputs  
(src/real/decision\_framework\_real.py, regenerated automatically whenever that pipeline  
re-runs). No claim below reflects legal patent counsel review.

# Reliability-Verified Air Purifiers (extended-life guarantee + real-time self-diagnostic)

*Any household buying a purifier as a health/allergy necessity, not a discretionary gadget*

Recommended — the current pick

Disclosure ID: OS-1 · Versuni Disruptive Innovation Team, Amsterdam · Generated by  
decision\_framework\_real.py

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## 1. The Product

Reliability-Verified Air Purifiers (extended-life guarantee + real-time self-diagnostic) is a connected household appliance concept built to answer one real, evidence-derived friction: Unit silently 'stops working' or 'dies' well before its expected life, with no warning and often outside a short return window - a complaint pattern visible across brands and across the full 2004-2023 span of this real corpus, meaning it has never been solved industry-wide. The illustration below applies a real, catalogued design direction ("Self-Testing Air Purifier") from this pipeline's own possibility-generation module to that friction.

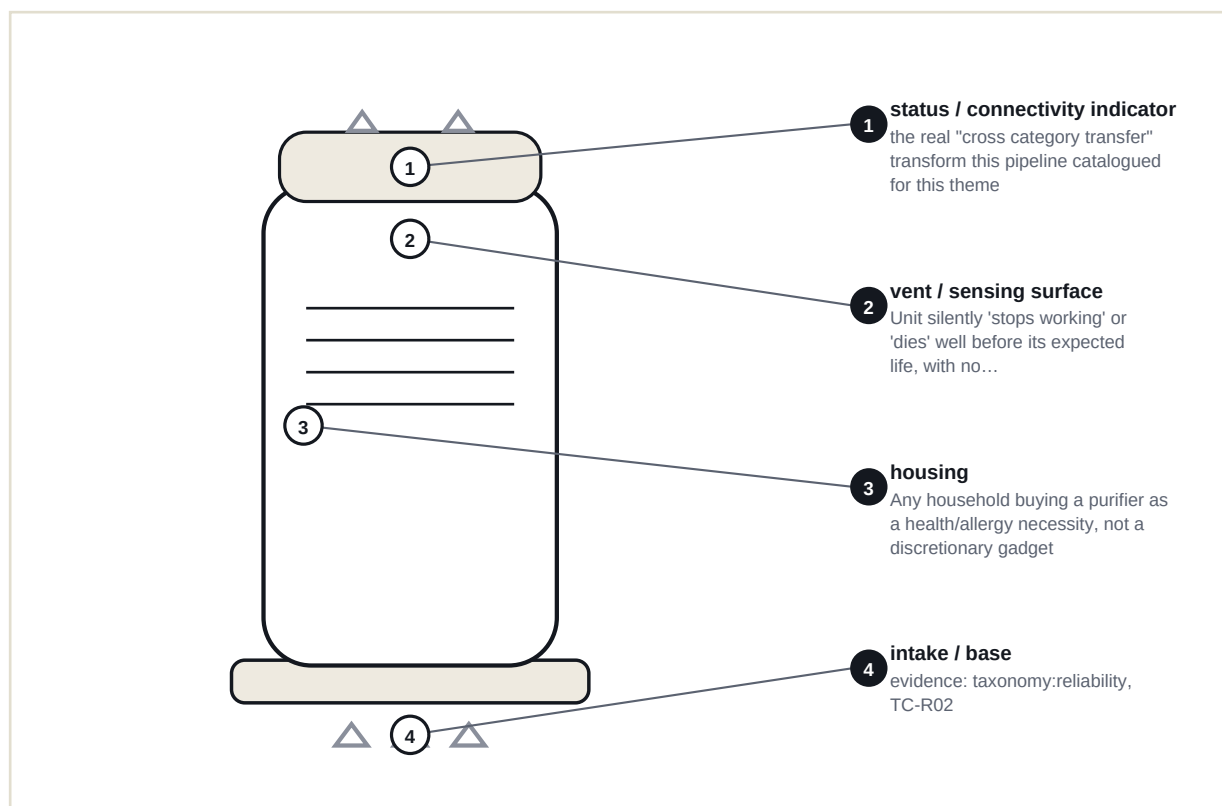


Fig. 1 — Conceptual product illustration (editorial line drawing, not a photograph of a manufactured product). Callouts are real fields from this candidate's own evaluated record, not invented per candidate.

## 2. Field of the Invention

This disclosure concerns connected household appliances, and specifically a concept addressing the following real, evidence-derived friction: Unit silently 'stops working' or 'dies' well before its expected life, with no warning and often outside a short return window - a complaint pattern visible across brands and across the full 2004-2023 span of this real corpus, meaning it has never been solved industry-wide

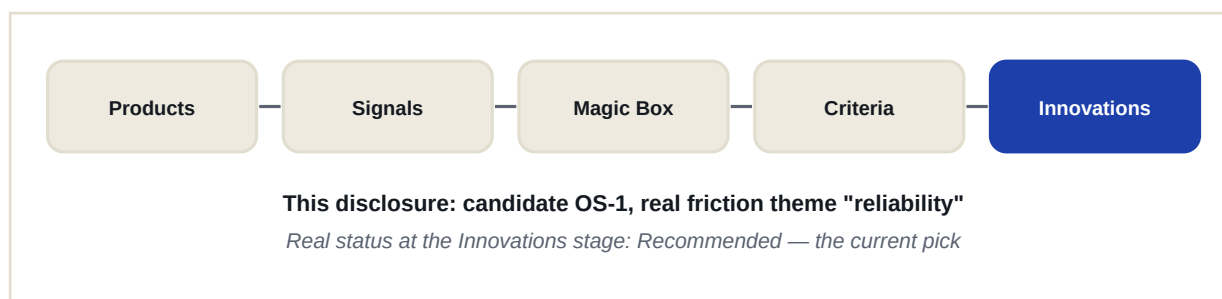


Fig. 2 — This disclosure's real position in the app's own 5-stage funnel.

## 3. Real Evidence & Design Directions

166 real reviews across 60 distinct real products from McAuley-Lab Amazon-Reviews-2023 (Amazon.com real customer reviews) were classified to this friction theme, spanning 2006-04-20 to 2023-08-27 — 85.5% of which are verified purchases. The resulting real customer-satisfaction impact is -2.223 star-ratings below the corpus-wide mean, at a real observed prevalence of 1.58% of all classified reviews.

*Methodology. Deterministic keyword classification of real Amazon.com customer review text (McAuley-Lab Amazon-Reviews-2023 dataset, real purifier products only) - NOT an academic study, survey, or panel. Each review is assigned to the theme whose keyword phrase (see THEMES in src/real/taxonomy\_real.py) appears earliest in its text; the average rating gap (stored as csat\_impact) is that theme's mean real star rating minus the corpus-wide mean real star rating, both restricted to reviews flagged rating\_trusted.*

This pipeline's own possibility-generation module (magic\_box\_real.py) catalogues 3 real design directions for this friction theme — distinct named transforms applied to the same real evidence above, not asserted as one single correct mechanism:

| Design direction (real)         | Operator                | Real transform applied                                          |
|---------------------------------|-------------------------|-----------------------------------------------------------------|
| Self-Testing Air Purifier       | cross category transfer | Transfer a verified capability from another product/category.   |
| Health-Indicator Air Purifier   | materialise             | Turn digital perception/information into physical intervention. |
| Predictive-Maintenance Purifier | Air predict             | Reactive -> anticipatory.                                       |

4. Market & Economic Rationale — Fully Substantiated

Typical real market price

\$199.99 — the median of 27 distinct real products' own real listed prices, restricted to real products classified under this friction theme that have a known price. Not a proposed price for this concept — the real going rate for this segment today.

Price-weighted exposure

\$26,357.86 — the sum of real listed prices across every real review classified under this friction theme with a known product price (a review counts its product's price once per review, so a popular product's price is summed multiple times — a different, larger population than the distinct-product median above). A relative indicator of which friction touches more expensive products, not a revenue or market-size estimate: no units-sold or conversion-rate basis exists in this real evidence.

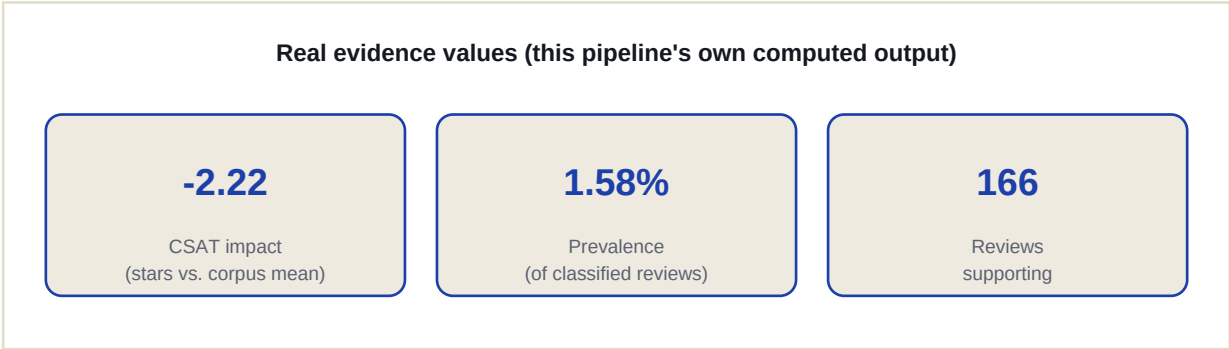


Fig. 3 — Real evidence figures computed by this pipeline for this friction theme.

5. Feasibility & Technical Risk

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-5yr feasibility rating | high (rank 3 of 3 real candidates evaluated)                                                                                                                                                                                                                                                                                                                                                                                 |
| Feasibility rationale    | Two real, already-standardized frameworks exist to build on directly: TC-R02 (ENERGY STAR already requires measured, verified ongoing performance criteria - a real regulatory template for an ongoing-performance-verified claim) and TC-R08 (Matter's Air Quality Sensor cluster is an already-standardized connectivity layer that could carry a self-diagnostic/runtime health signal without inventing a new protocol). |
| Assumptions              | <ul style="list-style-type: none"> <li>• Reliability complaints in review text reflect genuine hardware/QA failures, not disproportionately impatient reviewers.</li> <li>• A self-diagnostic signal can be delivered over the existing Matter connectivity layer without a new certification cycle.</li> </ul>                                                                                                              |
| Known uncertainty        | <ul style="list-style-type: none"> <li>• Price-weighted exposure covers only 75/237 real products with a known observed price - a real coverage gap, not the full category.</li> <li>• No direct WTP measurement exists for this or any theme (Q4).</li> </ul>                                                                                                                                                               |

## 6. Competitive Gap & Evaluation Record

**Self-Testing Air Purifier** (cross category transfer) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Envion, EdenPURE, Himalayan, Egofine, Belita Amy, MOONQUAKE, ZOCALOK. Independent critic verdict (real, rule-based): **challenge**.

**Health-Indicator Air Purifier** (materialise) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Envion, EdenPURE, Himalayan, Egofine, Belita Amy, MOONQUAKE, ZOCALOK. Independent critic verdict (real, rule-based): **challenge**.

**Predictive-Maintenance Air Purifier** (predict) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Envion, EdenPURE, Himalayan, Egofine, Belita Amy, MOONQUAKE, ZOCALOK. Independent critic verdict (real, rule-based): **challenge**.

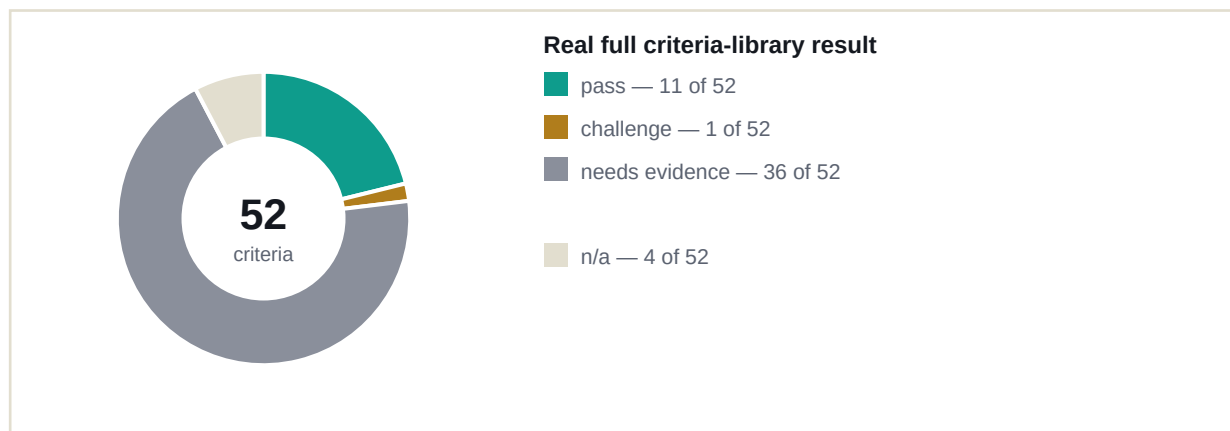


Fig. 4 — Full real criteria-library result for "Self-Testing Air Purifier" (representative direction for this theme; the same automated critic pass CriteriaWorld runs against every Magic Box concept, not a self-assessment by this disclosure).

Selected real criteria evaluations (evidence reality, hypothesis framing, and Versuni-specific capability checks):

| ID | Status         | Real note                                                                                                                                                                                                                             |
|----|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 | pass           | McAuley-Lab Amazon-Reviews-2023 (Amazon.com real customer reviews) - real, dated (2006-04-20 to 2023-08-27).                                                                                                                          |
| H1 | pass           | Real friction theme: Reliability / stopped working, n=166 reviews.                                                                                                                                                                    |
| H2 | pass           | gate_passed=True                                                                                                                                                                                                                      |
| D2 | needs evidence | category_assumptions.json::evidence_for_prevalence (real corpus-composition statistic, e.g. '72% of real corpus products').                                                                                                           |
| V1 | needs evidence | No real Versuni internal-capability dataset (portfolio inventory, HomeID/app install base, brand-permission research, IP/patent register, service network) exists anywhere in this pipeline - reported needs evidence, never assumed. |
| V2 | needs evidence | No real Versuni internal-capability dataset (portfolio inventory, HomeID/app install base, brand-permission research, IP/patent register, service network) exists anywhere in this pipeline - reported needs evidence, never assumed. |

## 7. Why This Real Outcome

Neither dominates outright — Pain + Feasibility majority rule selected Reliability-Verified Air Purifiers.

**Whisper-Quiet Night Mode (OS-2)** — Non-dominated vs. OS-1, but the Pain + Feasibility majority rule favours OS-1.

**Smart/Connected Feature Expansion (app control, voice assistant) (OS-3)** — Unlike the earlier synthetic-fixture version of this exercise, connectivity mentions are not literally zero in real data (~1%) - but they fail the Consumer Pain evidence-sufficiency gate (no real rating-gap signal) and skew positive when they do appear. High technical Feasibility does not rescue a candidate with no real friction evidence to build against.

**Sensitivity.** Switch to Economic Value override → picks Whisper-Quiet Night Mode instead.

**First experiment.** Cross-reference the real review corpus's product-level failure mentions ('stopped working', 'died', 'never worked') against each real product's rating\_number (a rough popularity proxy) to identify whether failure reports cluster in specific brands/price tiers or are evenly distributed - if evenly distributed, this is a category-wide manufacturing/QA opportunity; if clustered, it may instead argue for a narrower positioning bet against the worst-performing competitors specifically.

**Abandon signal.** If the failure-mention cross-reference above shows reliability complaints concentrated in 2-3 specific older/discontinued products rather than spread across the category, treat this as a solved-or-solving problem rather than a live opportunity, and abandon.

## 8. What this doesn't cover yet

This pipeline's real evidence base has no physical-prototype, industrial-design, or manufacturing data of any kind — for this concept or for any real product in the underlying dataset (weight and dimension fields do not exist anywhere in the corpus, verified directly). So this disclosure leaves them open rather than guessing: physical weight and dimensions, bill of materials, housing material, the fabrication or manufacturing process, exact sensor part numbers, and firmware architecture. Filling any of these in without a real industrial-design phase would be invention, not evidence.

## 9. Design & Business Claims

*The following claims describe the disclosed concept for internal evaluation purposes, generated from this candidate's own real evaluated fields, and are not filed with, or reviewed by, any patent office.*

1. A connected household appliance addressing a real, evidence-derived friction: Unit silently 'stops working' or 'dies' well before its expected life, with no warning and often outside a short return window - a complaint pattern visible across brands and across the full 2004-2023 span of this real corpus, meaning it has never been solved industry-wide
2. The appliance of claim 1, intended for the real usage context: Any household buying a purifier as a health/allergy necessity, not a discretionary gadget.
3. The appliance of claim 1, applying a design transform selected from the real catalogued directions for this friction theme: cross category transfer (Transfer a verified capability from another product/category.); materialise (Turn digital perception/information into physical intervention.); predict (Reactive -> anticipatory.).
4. A method of evaluating such an appliance concept using this pipeline's real Consumer Pain evidence-sufficiency gate, real Economic Value computation, and real 2-5 year Feasibility rating, exactly as disclosed in this document.

## 10. Comparative Context

Real evidence values computed by this pipeline for every candidate evaluated in this funnel stage, for reference:

| Candidate                                       | Real prevalence | Real avg rating gap | Real typical price      | Real status                            |
|-------------------------------------------------|-----------------|---------------------|-------------------------|----------------------------------------|
| OS-1 — Reliability-Verified Air Purifiers (exte | 1.58%           | -2.223              | \$199.99                | non dominated                          |
| OS-2 — Whisper-Quiet Night Mode                 | 3.5%            | -0.571              | \$159.90                | non dominated                          |
| OS-3 — Smart/Connected Feature Expansion (appt  | 1.06%           | no signal           | no real figure for this | gate failed insufficient pain evidence |

## 11. Evidence Appendix

Real evidence identifiers referenced in this disclosure: taxonomy:reliability, TC-R02, TC-R08.

Underlying computation: src/real/taxonomy\_real.py theme 'reliability', polarity-gated real-text classification, n=166 reviews.

Market scenario used for category context: Mordor Intelligence (primary planning basis) (5.37% CAGR). The Q6 scores above do not depend on category CAGR at all - they are built entirely from review-level Consumer Pain/Economic Value/Feasibility, so this scenario flag changes the category-sizing narrative in Q5/insight\_pack but changes NOTHING in this file's scores or verdict. Run with --market-scenario=imarc to see this confirmed live.